Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Find $\frac{3}{4} \times \frac{4}{5}$.

2. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

3. Find 25% of 32 stickers.

4. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

5. Find 0.25×0.04.

6. Expand $3(x + 6)$.

7. Find the surface area of a closed 1 by 4 by 5 cm box.

8. Find the opposite of -1.

9. Find the MAD of 3, 3, 9, 9.

10. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

11. A rectangle 9 cm by 4 cm is scaled by 2. Find new dimensions and area.

12. A drawing has area 36 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

13. A map uses 1 cm for 9 km. A route is 4.5 cm on the map. Find actual distance.

14. A proportional table includes (x, y) = (4, 36). Find k, and find y when x = 7.

15. A rental costs \$9 to start plus \$4 per hour. Write cost c for h hours. Is the relationship proportional?

16. A pump fills 9 liters in 4 minutes. At a constant rate, write L after t minutes and find the amount after 8 minutes.

17. A circle has diameter 18 cm. Find circumference in terms of π and using 3.14.

18. A circle has radius 4 cm. Find area in terms of π and using 3.14.

CHALLENGE / Explain your plan and check the result

C1. A rectangular plaza is 12 m by 8 m. A circular fountain of diameter 4 m occupies part of it. Using 3.14, find the remaining paving area. Paving costs \$10 per square meter, and a10% discount applies to the paving bill. Find the discounted cost.

C2. A print shop charges Plan A: \$6 setup plus \$0.20 per page, and Plan B: \$0.35 per page. Find both prices for 20 and 60 pages. Find the page count where they agree. Decide whether either plan is proportional to page count and justify.